

Tabular supervised problem



1. Serving-latency budget critical?
(FMs cost 4-5 orders of magnitude more per row)

yes

GBDT family
(XGBoost / LightGBM / CatBoost)
distillation can recover part
of the FM edge (AI-2)

no



2. Architectural limits hit?
(>10 classes; context cap)

yes

GBDT or DL
(task-matched; see reversals)

no



3. Intrinsic interpretability required?
(regulation)

yes

GBDT
(+ SHAP on the side)

no



4. No active constraint → FOUNDATION MODEL FIRST
TabPFN (mature) · TabFM (frontier, v1.0.1 caveats)
median normalized regret 0.000 @14 models (LODO)